

Lesson 8-1: Understand Equations and Their Solutions

Connect & apply • 8 tasks • Prepares for Standard 6.EE.B.5

Name: _____ Date: _____ Period: _____

Show complete thinking. Label units. Use an equation, sketch, or model. Accuracy before speed.

TASK 1 • FOUNDATION 1

Evaluate $3x + 2$ when $x = 4$.

TASK 2 • FOUNDATION 2

Find the missing number: $7 + 5 = __ + 4$.

TASK 3 • FOUNDATION 3

Write an equation for a number n plus 6 is 15.

TASK 4 • FOUNDATION 4

Does $x = 5$ make $2x + 1 = 11$ true? Explain.

One strategy or model I used: _____

My next step: ☐ Explain my steps ☐ Try another problem ☐ Review with a partner

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Show complete thinking. Label units. Use an equation, sketch, or model. Accuracy before speed.

TASK 5 · QUICK CHECK A

Is $x = 4$ a solution of $3x = 12$? Write an equation for "three times a number is 15."

TASK 6 · QUICK CHECK B

Is $x = 5$ a solution of $4x = 20$? Write an equation for "five times a number is 35."

TASK 7 · QUICK CHECK C

Is $x = 3$ a solution of $2x + 1 = 8$? Write an equation for "a number increased by 9 is 21."

TASK 8 · TRANSFER & EXTEND

Which of 2, 4, or 6 is a solution of $3x - 2 = 10$?

One strategy or model I used: _____

My next step: ☐ Explain my steps ☐ Try another problem ☐ Review with a partner

Lesson 8-2: Write and Solve Equations Using Addition or Subtraction

Connect & apply · 8 tasks · Prepares for Standard 6.EE.B.7

Name: _____ Date: _____ Period: _____

Show complete thinking. Label units. Use an equation, sketch, or model. Accuracy before speed.

TASK 1 • FOUNDATION 1

Find the missing number: $\underline{\quad} + 7 = 19$.

TASK 2 · FOUNDATION 2

Compute $5.2 - 1.85$.

TASK 3 · FOUNDATION 3

Which operation undoes subtracting 8?

TASK 4 · FOUNDATION 4

Check $x = 9$ in $x + 6 = 15$. State the left-side value.

One strategy or model I used: _____

My next step: ☐ Explain my steps ☐ Try another problem ☐ Review with a partner

Lesson 8-2: Write and Solve Equations Using Addition or Subtraction

Connect & apply · 8 tasks · Prepares for Standard 6.EE.B.7

Name: _____ Date: _____ Period: _____

Show complete thinking. Label units. Use an equation, sketch, or model. Accuracy before speed.

TASK 5 · QUICK CHECK A

$m - 11 = 25$, so $m = ?$ What undoes "+ 8"?

TASK 6 · QUICK CHECK B

$m - 14 = 31$, so $m = ?$ What undoes "+ 12"?

TASK 7 · QUICK CHECK C

$y + 18 = 40$, so $y = ?$ What undoes "- 7"?

TASK 8 · TRANSFER & EXTEND

A jar lost 17 marbles and now holds 48. Write and solve an equation for the original count.

One strategy or model I used: _____

My next step: ☐ Explain my steps ☐ Try another problem ☐ Review with a partner

Lesson 8-3: Write and Solve Equations Using Multiplication or Division

Connect & apply · 8 tasks · Prepares for Standard 6.EE.B.7

Name: _____ Date: _____ Period: _____

Show complete thinking. Label units. Use an equation, sketch, or model. Accuracy before speed.

TASK 1 • FOUNDATION 1

Compute 9×7 .

TASK 2 · FOUNDATION 2

Compute $12.6 \div 3$.

TASK 3 • FOUNDATION 3

Which operation undoes multiplying by 6?

TASK 4 · FOUNDATION 4

Compute $12 \div \frac{3}{4}$.

One strategy or model I used: _____

My next step: ☐ Explain my steps ☐ Try another problem ☐ Review with a partner

Lesson 8-3: Write and Solve Equations Using Multiplication or Division

Connect & apply · 8 tasks · Prepares for Standard 6.EE.B.7

Name: _____ Date: _____ Period: _____

Show complete thinking. Label units. Use an equation, sketch, or model. Accuracy before speed.

TASK 5 • QUICK CHECK A

$6x = 42$, so $x = ?$ $x \div 5 = 9$, so $x = ?$ What undoes " $\times 4$ "?

TASK 6 • QUICK CHECK B

$8x = 56$, so $x = ?$ $x \div 4 = 7$, so $x = ?$ What undoes " $\times 9$ "?

TASK 7 • QUICK CHECK C

$5x = 45$, so $x = ?$ $x \div 3 = 12$, so $x = ?$ What undoes " $\div 6$ "?

TASK 8 · TRANSFER & EXTEND

Tickets cost the same amount. Nine tickets cost \$63. Write and solve an equation for one ticket.

One strategy or model I used: _____

My next step: ☐ Explain my steps ☐ Try another problem ☐ Review with a partner

Lesson 8-4: Write and Represent Inequalities

Connect & apply • 8 tasks • Prepares for Standard 6.EE.B.8

Name: _____ Date: _____ Period: _____

Show complete thinking. Label units. Use an equation, sketch, or model. Accuracy before speed.

TASK 1 • FOUNDATION 1

Which symbol means greater than or equal to?

TASK 2 • FOUNDATION 2

Write an inequality for a number n that is at least 12.

TASK 3 • FOUNDATION 3

On a number line marked in ones, what number is two steps left of 1?

TASK 4 • FOUNDATION 4

Does 7 satisfy $n < 5$? Explain.

One strategy or model I used: _____

My next step: ☐ Explain my steps ☐ Try another problem ☐ Review with a partner

Lesson 8-4: Write and Represent Inequalities

Connect & apply • 8 tasks • Prepares for Standard 6.EE.B.8

Name: _____ Date: _____ Period: _____

Show complete thinking. Label units. Use an equation, sketch, or model. Accuracy before speed.

TASK 5 • QUICK CHECK A

Write an inequality for "you must be at least 48 inches tall." Is 50 a solution?

TASK 6 • QUICK CHECK B

Write an inequality for "you may bring at most 3 guests." Is 3 a solution?

TASK 7 • QUICK CHECK C

Write an inequality for "the ride requires more than 40 inches." Is 40 a solution?

TASK 8 • TRANSFER & EXTEND

A bus holds no more than 44 riders and already has 31. Write an inequality for how many more can board.

One strategy or model I used: _____

My next step: ☐ Explain my steps ☐ Try another problem ☐ Review with a partner

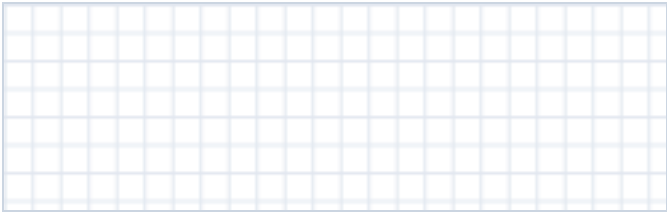
Lesson 8-5: Understand Inequalities and Their Solutions

Connect & apply · 8 tasks · Prepares for Standard 6.EE.B.5

Name: _____ Date: _____ Period: _____

Show complete thinking. Label units. Use an equation, sketch, or model. Accuracy before speed.

TASK 1 · FOUNDATION 1
Describe the graph of $x > 3$ on a number line.



TASK 2 · FOUNDATION 2
For $x \leq 4$, is the circle at 4 open or closed?

TASK 3 · FOUNDATION 3
Which of 2, 5, 8 satisfy $x \geq 5$?

TASK 4 · FOUNDATION 4
Which is greater: -2 or -6?

One strategy or model I used: _____

My next step: ☐ Explain my steps ☐ Try another problem ☐ Review with a partner

Lesson 8-5: Understand Inequalities and Their Solutions

Connect & apply · 8 tasks · Prepares for Standard 6.EE.B.5

Name: _____

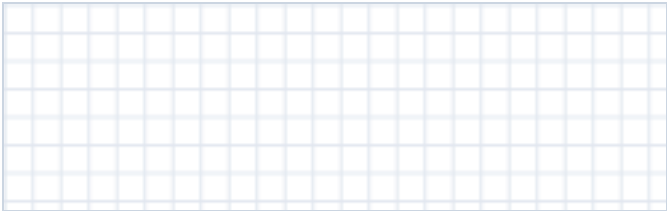
Date: _____

Period: _____

Show complete thinking. Label units. Use an equation, sketch, or model. Accuracy before speed.

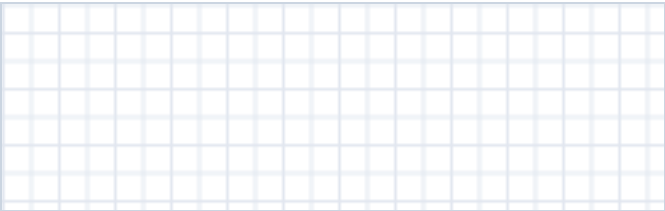
TASK 5 · QUICK CHECK A

Graph $x \geq 3$. Open circle or closed? Is 3 itself a solution?



TASK 6 · QUICK CHECK B

Graph $x > 5$. Open circle or closed? Is 5 itself a solution?



TASK 7 · QUICK CHECK C

Graph $x \leq -2$. Open circle or closed? Is -2 itself a solution?



TASK 8 · TRANSFER & EXTEND

Which of -1 , 0 , 4 , and 4.5 satisfy $x \geq 4$?

One strategy or model I used: _____

My next step: ☐ Explain my steps ☐ Try another problem ☐ Review with a partner